

2 BIT ARITHMETIC LOGIC UNIT (ALU)

A Hardware Digital System Design Project by
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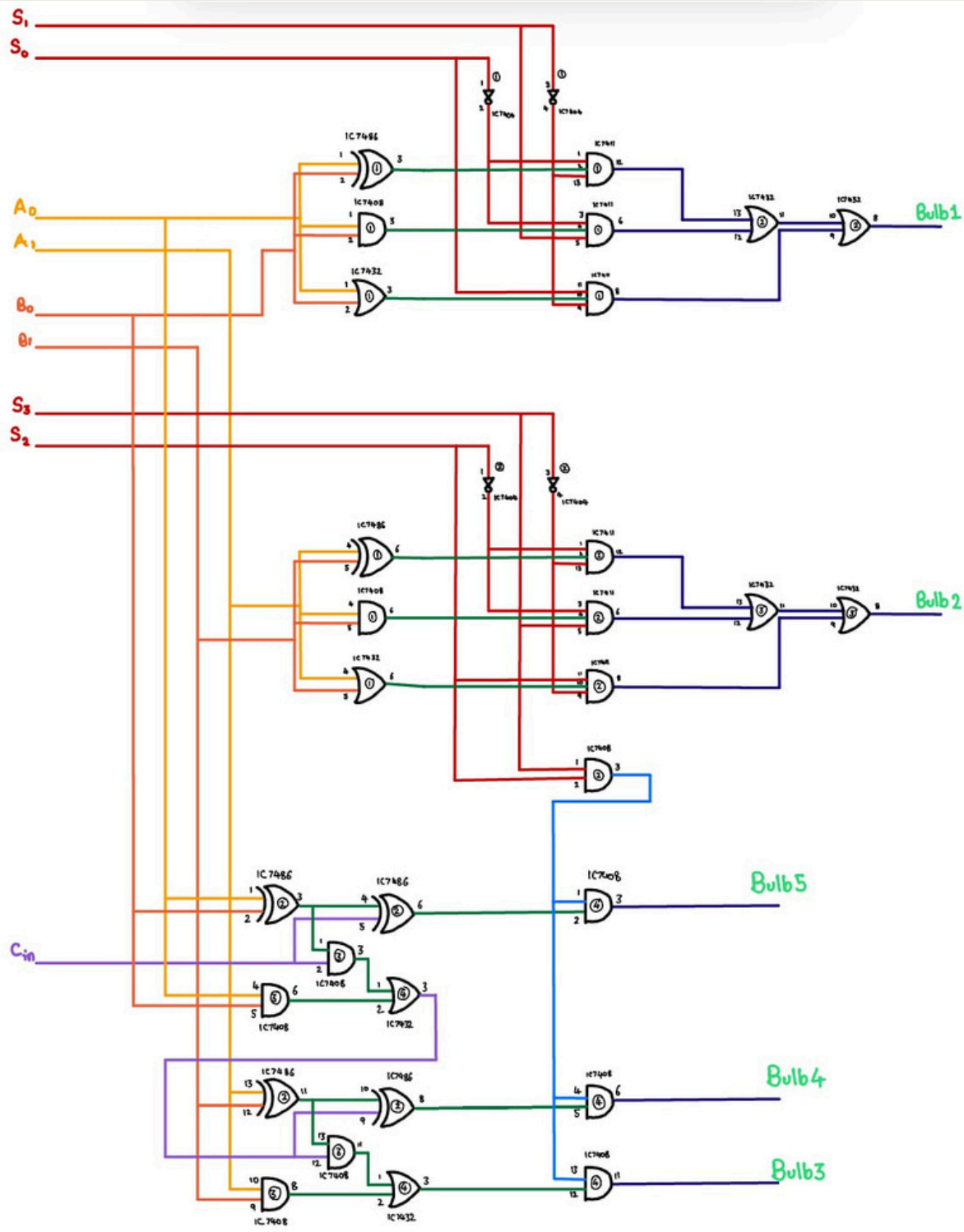
OBJECTIVE

An Arithmetic Logic Unit (ALU) is a fundamental building block of a computer's central processing unit (CPU). It performs arithmetic and logical operations on binary data. A 2-bit ALU is a simplified version of an ALU that operates on two 2-bit binary inputs, A1 A0 and B1 B0, and produces a 2-bit output based on the selected operation. The operation is determined by four select lines, S1 and S0, which act as control signals for A0 and B0 and S2, S3 for A1 and B1 respectively.

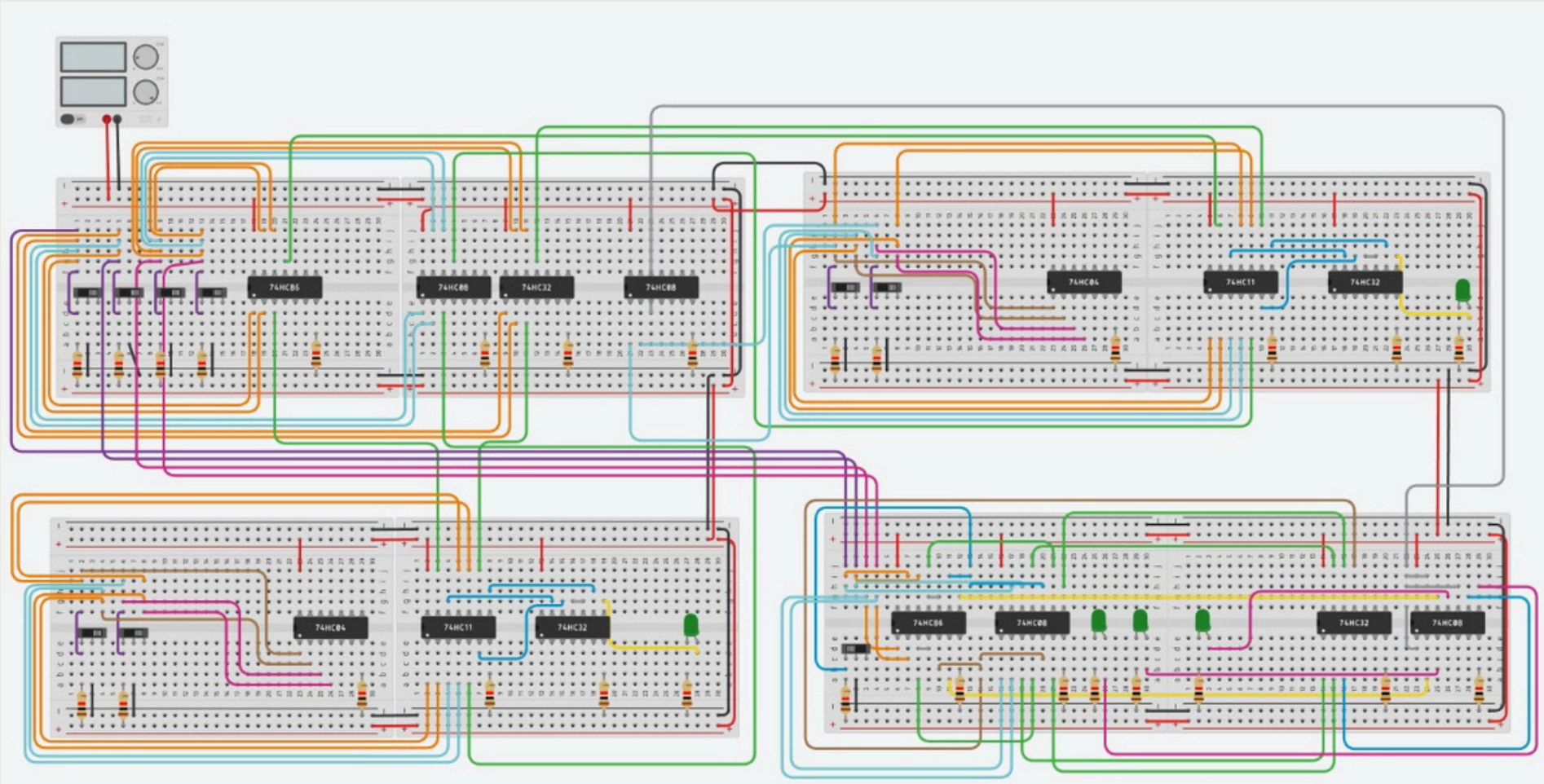
COMPONENTS

1. Power Supply (3 AA batteries)
2. Switch
3. Slide Switches (9)
4. LEDs (Green) (5)
5. Breadboard (4)
6. Connecting Wires
7. XOR gate (**IC7486**) (2)
8. AND gate (**IC7408**) (4)
9. OR gate (**IC7432**) (4)
10. NOT gate (**IC7404**) (2)
11. 3 Input AND gate (**IC7411**) (2)
12. Resistor (**1k Ohm**) (28)

CIRCUIT DIAGRAM



CIRCUIT DIAGRAM (DIGITAL)



TRUTH TABLE

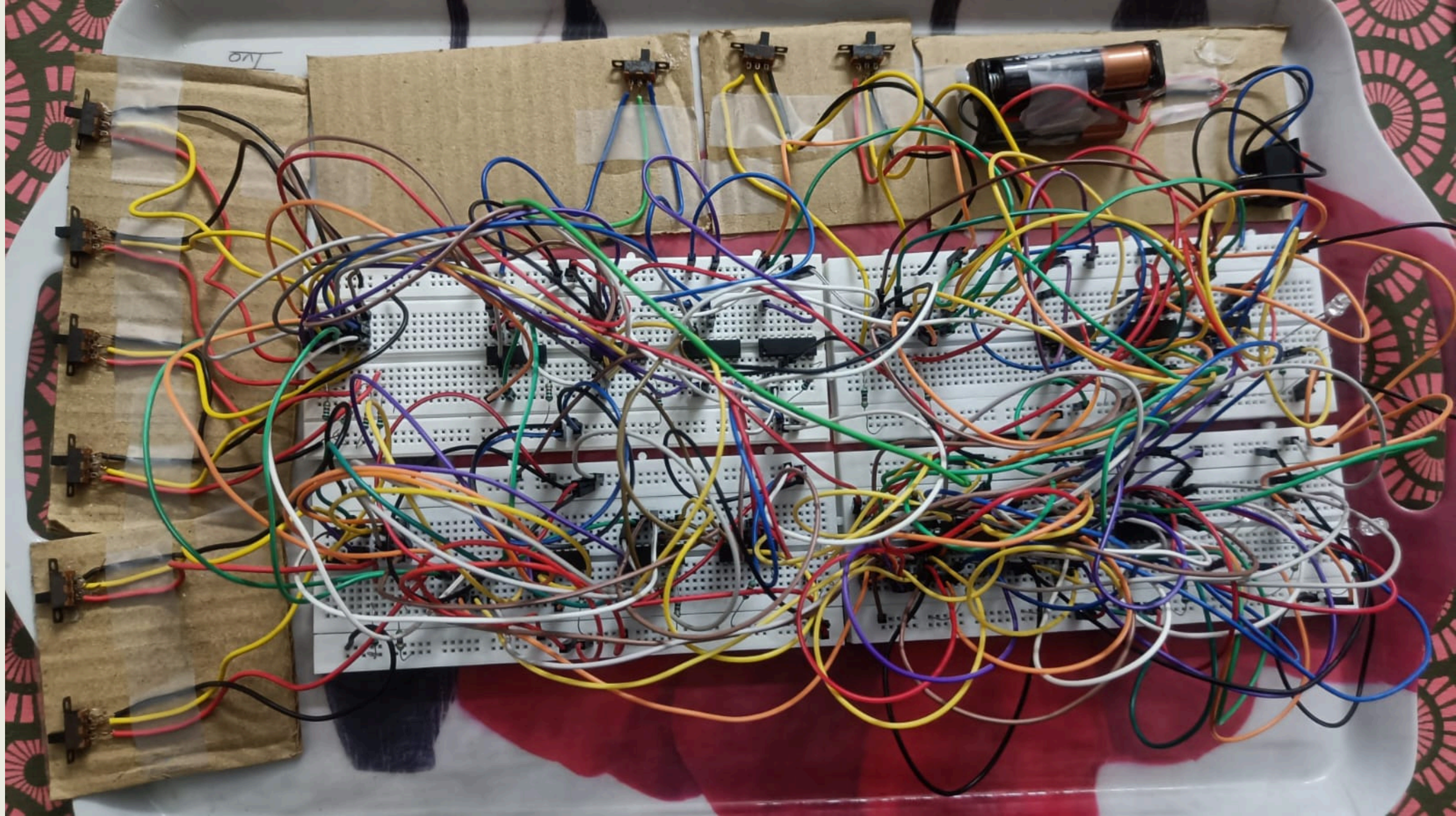
A1	A0	B1	B0	Cin	S0	S1	S2	S3	Bulb1	Bulb2	Bulb3	Bulb4	Bulb5
0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	0	0	1	0	0	0	0
0	1	0	0	0	0	0	0	0	1	0	0	0	0
0	1	0	1	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	1	0	0	0	0	0	0	0
0	0	0	1	0	0	1	0	0	0	0	0	0	0
0	1	0	0	0	0	1	0	0	0	0	0	0	0
0	1	0	1	0	0	1	0	0	1	0	0	0	0
0	0	0	0	0	1	0	0	0	0	0	0	0	0
0	0	0	1	0	1	0	0	0	1	0	0	0	0
0	1	0	0	0	1	0	0	0	1	0	0	0	0
0	1	0	1	0	1	0	0	0	1	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	1	0	0	0	0	0	0	0	1	0	0	0
1	0	0	0	0	0	0	0	0	0	1	0	0	0
1	0	1	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	1	0	0	0	0	0
0	0	1	0	0	0	0	0	1	0	0	0	0	0
1	0	0	0	0	0	0	0	1	0	0	0	0	0
1	0	1	0	0	0	0	0	1	0	1	0	0	0
0	0	0	0	0	0	0	1	0	0	0	0	0	0
0	0	1	0	0	0	0	1	0	0	1	0	0	0
1	0	0	0	0	0	0	1	0	0	1	0	0	0
1	0	1	0	0	0	0	1	0	0	1	0	0	0
0	0	0	0	0	1	1	1	1	0	0	0	0	0
0	0	0	1	0	1	1	1	1	0	0	0	0	1
0	0	1	0	0	1	1	1	1	0	0	0	1	0
0	0	1	1	0	1	1	1	1	0	0	0	1	1
0	1	0	0	0	1	1	1	1	0	0	0	0	1
0	1	0	1	0	1	1	1	1	0	0	0	1	0

0	1	1	0	0	1	1	1	1	0	0	0	1	1
0	1	1	1	0	1	1	1	1	0	0	1	0	0
1	0	0	0	0	1	1	1	1	0	0	0	1	0
1	0	0	1	0	1	1	1	1	0	0	0	1	1
1	0	1	0	0	1	1	1	1	0	0	1	0	0
1	0	1	1	0	1	1	1	1	0	0	1	0	1
1	1	0	0	0	1	1	1	1	0	0	0	1	1
1	1	0	1	0	1	1	1	1	0	0	1	0	0
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0	0	0	0	1	1	1	1	1	0	0	0	0	1
0	0	0	1	1	1	1	1	1	0	0	0	1	0
0	0	1	0	1	1	1	1	1	0	0	0	1	1
0	0	1	1	1	1	1	1	1	0	0	1	0	0
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0	1	0	1	1	1	1	1	1	0	0	0	1	1
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1	1	1	1	1	1	1	1	1	0	0	1	1	1

SIMULATION

[https://www.tinkercad.com/things/85IWxWIZ1Ph/editel?
returnTo=%2Fdashboard&sharecode=C0e4D2rIfS89xsmvZTMxLiRMtTyORuYCjvhItnnf1Fw](https://www.tinkercad.com/things/85IWxWIZ1Ph/editel?returnTo=%2Fdashboard&sharecode=C0e4D2rIfS89xsmvZTMxLiRMtTyORuYCjvhItnnf1Fw)

SETUP



REFERENCES

<https://eater.net/8bit/alu>

<https://medium.com/@vaibhav.shingde22/2-bit-alu-logic-operations-9110f1e89ea0>

